INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

MULTIMETER, RUGGEDIZED

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID), describes a Ruggedized Multimeter. This CID is meant as a minimum requirement for the Ruggedized Multimeter in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

2. CLASSIFICATION. The multimeter shall conform to the following types:

2.1 Type I shall include the multimeter and stray voltage test leads.

2.2 Type II shall be the multimeter without the stray voltage test leads.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and be bid sample tested within the accuracy, limits, and specifications herein.

3.1 Electrical power sources. The only power from which the multimeter shall operate is the power supplied by the battery contained within the multimeter.

3.1.1 Battery. One 9V battery, National Electronic Distributors Association (NEDA) type 1604, 9V alkaline, or equal, shall be used within the multimeter. In voltage or current measurement functions (under worse case battery drain conditions), the battery shall be capable of sustaining 600 hours of operation at 25°C and 300 hours of operation at -20°C or 55°C. In resistance or diode test functions (under worst case battery conditions), the battery shall be capable of sustaining 300 hours of operation at 25°C and 250 hours of operation at -20°C or 55°C. The battery may be shipped installed.

3.1.2 Battery test. A display indication shall be provided to show marginal performance of the multimeter due to low battery potential.

3.2 Temperature requirements.

3.2.1 Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -51°C to +71°C.

3.2.2 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of -40°C to +55°C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Bldg 171, Kelly AFB, TX 78241-6524.

AMSC N/A FSC 6625

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3.3 Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours.

3.4 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effect of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

3.5 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

3.6 Performance characteristics.

3.6.1 Accuracy. Unless otherwise specified, all electrical specifications in paragraph are from 18°C to 28°C with up to 95 percent humidity, for one year after calibration. At temperature less than 18°C or greater than 28°C, the temperature coefficient shall be less than 0.1 times the applicable electrical specification per °C. Accuracy is specified as +/- (percent of reading plus number of digits).

3.6.2 DC voltage.

3.6.2.1 Ranges, resolution, and accuracy. As specified in Table I or better.

TABLE I. DC Voltage.

RANGE	RESOLUTION	ACCURACY
320.0 mV	100 uV	+/- (0.2% +1)
3.200 V	1 mV	"
32.00 V	10 mV	"
320.0 V	100 mV	"
1000 V	1V	"

3.6.2.2 Input impedance. The input impedance shall be greater than 9.9 Megohms, paralleled by less than 100 picofarads, on any range.

3.6.2.3 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1,000 Vrms on all ranges, except 500 Vrms continuously on the 300.0 mV range.

3.6.2.4 Normal mode rejection ratio. Normal mode noise rejection shall not be less than 60 decibels (dB) at 50 and 60 Hz (+/-0.15%).

3.6.2.5 Common mode rejection ratio. Common mode rejection, with a 1,000 Ohms unbalanced in the low lead, shall not be less than 120 decibels (dB) at 50 and 60 Hz (+/-0.15%).

3.6.3 AC Voltage.

3.6.3.1 Range, resolution, and accuracy. As specified in Table II or better.

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TABLE II - AC Voltage.

RANGE	RESOLUTION	ACCURACY		
		40 Hz to 2 KHz	2 KHz to 10 KHz	10 KHz to 30 KHz
320.0 mV	100 uV	+/-(0.5%+3)	+/-(2%+3)	+/-(4%+10)
3.200 V	1 mV	"	"	"
32.00 V	10 mV	"	"	"
320.0 V	100 mV	"	"	"
1000 V	1V	+/-(1%+3)	+/-(3%+3)	Not specified

3.6.3.2 Input impedance. The input impedance shall be greater than 9.9 Megohms, paralleled by less than 100 picofarads, on any range.

3.6.3.3 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1,000 Vrms on all ranges; except 500 Vrms continuously on the 300 mV range. The multimeter shall be capable of withstanding an AC Volt-Hertz product up to 107.

3.6.3.4 Common mode rejection ratio. Common mode rejection, with a 1,000 ohms unbalance in the low lead, shall not be less than 60 decibels (dB) at 50 and 60 Hz (+/-0.15%).

3.6.4 Resistance.

3.6.4.1 Ranges, resolution and accuracy. As specified in Table III or better.

TABLE III. Resistance.

RANGE	RESOLUTION	ACCURACY
320.0 ohm	0.1 ohm	+/- (0.3%+2)
3.200 Kohm	1 ohm	+/- (0.2%+1)
32.00 Kohm	10 ohm	"
320.0 Kohm	100 ohm	"
3.200 Mohm	1 Kohm	"
32.00 Mohm	10 Kohm	+/- (1%+1)
32.00 nS	0.01nS	+/- (2%+10)

3.6.4.2 Humidity. The 32.00 nS range accuracy specification shall be limited to 70% maximum relative humidity from 28°C to 55°C.

3.6.4.3 Maximum current through unknown resistance. The multimeter shall not provide more than 1.1 mA on any range.

3.6.4.4 Maximum open terminal voltage. The open terminal voltage shall not be greater than 3.0 volts on any range.

3.6.4.5 Maximum on scale test voltage. The multimeter shall be capable of allowing in-circuit resistance measurements. It shall not apply greater than 450 mV at full scale on the following ranges: 320 ohms, 3.2 Kohm, 32 Kohm, 320 Kohm, and 3.2 Mohm.

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3.6.4.6 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 500 Vrms on all ranges.

3.6.5 Diode and continuity test.

3.6.5.1 Continuity test. The multimeter shall provide a continuous audible tone (or equivalent) for test resistance below 100 ohms. Forward biased (0.15 volts or greater) diode or transistor junctions shall inhibit continuous tone. Any deviation requires Air Force approval.

3.6.5.2 Diode test. The multimeter shall be capable of displaying diode or transistor junction voltage drops to at least 2.000 volts. A momentary audible tone shall indicate a forward biased junction, and the absence of sound shall indicate a reversed biased or open junction.

3.6.5.3 Maximum test current. The maximum diode or continuity test current shall be 1.1 mA.

3.6.5.4 Maximum open terminal voltage. The open terminal diode or continuity test voltage shall not be greater than 3.5 volts.

3.6.5.5 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 500 Vrms on all ranges.

3.6.6 DC and AC current.

3.6.6.1 Ranges, resolution, accuracy, and burden voltage. As specified in Table IV or better.

TABLE IV. DC and AC Current.

RANGE	RESOLUTION	DC ACCURACY	AC ACCURACY 40 Hz TO 1 KHz	MAX BURDEN (V)
320.0 uA	0.1 uA	+/-(0.75%+2)	+/-(1.5%+2)	160 mV
3200 uA	1 uA	"	"	1.6 V
32.00 mA	10 uA	"	"	180 mV
320.0 mA	100 uA	"	"	1.8 V
10.00 A	10 mA	"	"	0.5 V

3.6.6.2 Overload. The multimeter shall be capable of continuously withstanding, without damage, 0.4 Vrms on all ranges; except the 10 A range shall be capable of withstanding 10 Vrms continuously and 20 A for 30 seconds. All current inputs shall be fused or protected to interrupt overload currents, up to 10,000 A, from accidental voltage overloads in circuits up to 600 volts.

3.6.7 Controls and visual display.

3.6.7.1 Ranges and function control. All input functions (voltage, current, resistance, and diode/continuity test) shall be controlled with one rotary switch. Ranging shall be automatic (with a manual override) in DC and AC voltage from the 3 V range to the 1,000 V range, and in resistance from the 300 ohm range to the 30 Mohm range. Ranging shall be automatic (with a manual override) in DC and AC current between 30 mA and 300 mA; and between 300 uA and 3,000 uA.

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3.6.7.2 Readout display. The readout display shall include a liquid crystal display (LCD) with a digital readout with 3,200 counts of resolution, an analog bar-graph (or equivalent) readout with at least 30 segments of resolution, and auto-zeroing. The digital display shall update at least 2 times per-second and the analog display shall update at least 20 times per-second.

3.6.7.3 Automatic polarity. The polarity of the measured value shall be automatically indicated on the digital and analog displays.

3.6.7.4 Response time. Response time is defined as the time interval between an abrupt change of input and the time the new value is displayed within specified accuracy, including the worst case time to change ranges in the auto range mode.

3.6.7.5 Direct current or voltage response. Direct current or voltage response shall be less than 2 seconds.

3.6.7.6 Alternating current or voltage response. Alternating current or voltage response time shall be less than 4 seconds.

3.6.7.7 Resistance response time. Resistance response time shall be less than 2 seconds on all ranges except that in the nano-Siemen range or the lowest and highest resistance ranges, it shall not be more than 6 seconds.

3.6.7.8 Reading hold. The multimeter shall be capable of holding a reading in the display, for the operator to view, after the test leads have been removed from the device under test.

3.6.8 Accessories.

3.6.8.1 Stray voltage test leads. (Optional) 1 Kohms +/- 20% across input circuit. Rated to 500 VDC or VAC, without damage to probes.

3.6.8.2 Test leads. Two test leads (one red, one black) shall be supplied with the multimeter. The test leads shall be at least 1 meter long, rated to withstand at least 1,500 volts, and have right angle safety connectors.

3.6.8.3 Meter case. Non-metallic case, ruggedized to withstand harsh environment normally found in flight line applications.

3.6.9 Maintenance and calibration accessories. If required, the multimeter shall be provided with all necessary maintenance and calibration accessories. Not more than one circuit board extender or patch cable shall be required to permit maintenance of the modules and plug-in circuit cards.

4. QUALITY ASSURANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Regulatory requirements. The offeror/contractor is encouraged to use the recovered materials in accordance with Public Law 94-580 and paragraph 23.403 of the Federal Acquisition Regulation (FAR) to the maximum extent practical.

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5. PACKAGING.

5.1 Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified in the inch/pound units. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6.2 The contracting officer has the option of accepting or rejecting the product.

6.3 Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

6.4 Safety and environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800.

6.5 Government Standards.

MIL-PRF-28800 Test Equipment for use with Electrical and Electronic Equipment, General Specification for

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

6.6 Acceptable manufacturers. The following brand name, model and national stock number are known to be acceptable and current manufacturer is listed below:

Manufacturer	Model	National Stock Number
Fluke	80258	6625-01-266-3494
Procurement Document	A-A-50734	6625-01-079-1762

6.7 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part number system for CID A-A-50734.

AA50734-I denotes Type I
AA50734-II denotes Type II

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MILITARY INTEREST:

Custodian:
Air Force 99
Army MI
Navy SH

Reviewer:
Army - CR
DLA - ES
OSD/SO

CIVIL AGENCY COORDINATING ACTIVITY
GSA-FSS

Preparing Activity:
Air Force - 82

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